

Applying the criteria for congruence

1 Which of the following are criteria for proving two triangles are congruent? Tick **3** correct answers

☐ AAA

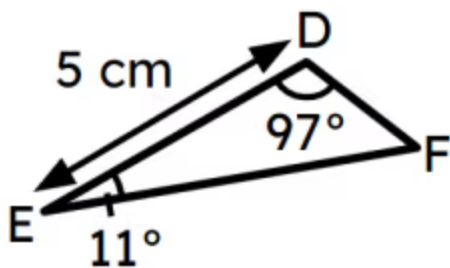
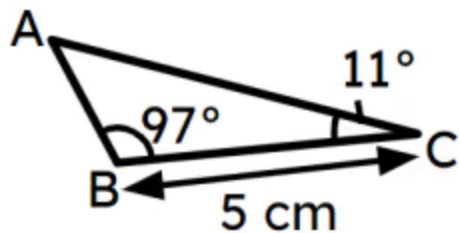
☒ ASA

☒ SAS

☐ SSA

☒ RHS

2 Triangle ABC and triangle DEF are congruent by _____. Tick **1** correct answer



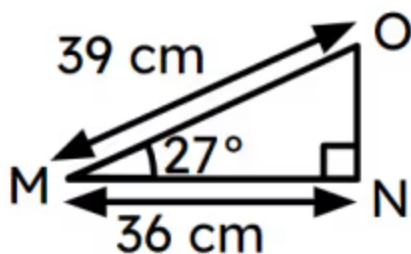
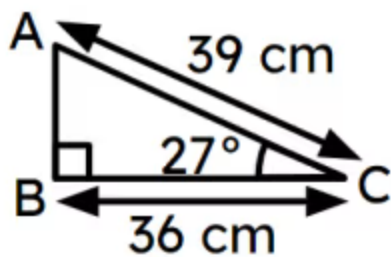
☒ ASA

☐ RHS

☐ SAS

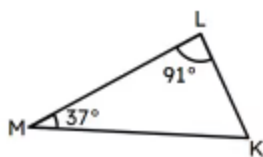
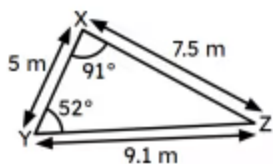
☐ SSS

- 3 There is insufficient information to prove triangle ABC is congruent to triangle MNO by _____. Tick **1** correct answer



- ☒ ASA
☐ RHS
☐ SAS
☐ SSS

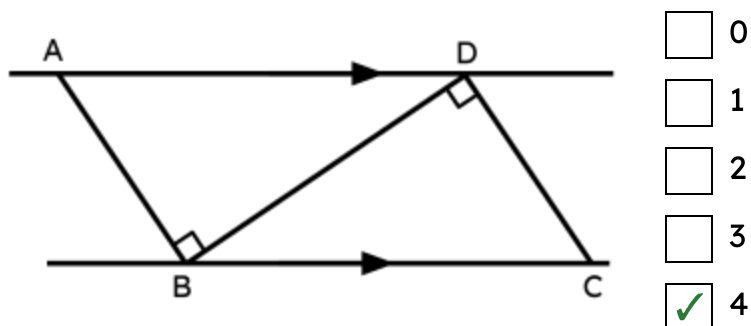
- 4 Triangle KLM is _____ to triangle XYZ. Tick **1** correct answer



- ☐ definitely congruent
☐ definitely incongruent
☒ potentially congruent



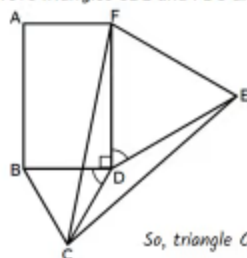
- 5 If you know that ABCD is a parallelogram, then _____ criteria for congruent triangles (ASA, RHS, SAS and SSS) can be used to show ABD and BDC are congruent. Tick 1 correct answer



- 6 Matching a, b, c and d to the correct statements to complete this congruence proof. Write the correct letter in each box

ABDF is a rectangle. BCD and DEF are equilateral triangles.

Prove triangles CDE and FDC are congruent.



a is a common edge to both triangles

$DF = \mathbf{b}$ as they are edges of the same equilateral triangle

$\mathbf{c} = 90 + 60 = 150^\circ$

$\mathbf{d} = 360 - (60 + 90 + 60) = 150^\circ$

Therefore, $\angle CDF = \angle CDE$

So, triangle CDE is congruent to triangle FDC by SAS.

a	a
b	b
c	c
d	d

d	angle CDE
c	angle CDF
b	DE
a	CD

